# On Embedding 2-Dimensional Toroidal Grids into de Bruijn Graphs with Clocked Congestion One

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**Abstract.** For integers $m, d, D$ with $m \geq 3, d \geq 2$, and $D \geq 2$, let $T(m)$ be a 2–dimensional quadratic toroidal grid with side length $m$ and let $B(d, D)$ be the base $d$, dimension $D$ de Bruijn graph; assume that $|T(m)| = |B(d, D)|$. The starting point for our investigations is the observation that, for $m, D$ even, embeddings $f : T(m) \to B(d, D)$ with load 1, expansion 1, and dilation $D/2$ can easily be found (and have previously been described in the literature). In the present paper, we pose the question whether or not there exist embeddings $f : T(m) \to B(d, D)$ with these properties *and with clocked congestion 1.* We prove results implying a positive answer to this question when $d$ is greater than two. For $d = 2$, we do not have a complete answer, but present partial results.

*Key words: de Bruijn graphs, toroidal grids, graph embeddings, clocked congestion, dilation, interconnection networks, parallel computers*

## 1 Introduction

The problem of embedding one interconnection network into another one is of significant importance in the context of parallel computers. Among the network topologies which have been proposed and investigated in the literature, toroidal grids (briefly: tori) and de Bruijn graphs are among the most popular ones; see, e.g., [1, 3, 4, 6, 7, 8, 9]. In the present paper, we deal with the problem of constructing "good" embeddings of a 2-dimensional quadratic torus into a de Bruijn graph $B(d, D)$ with the same number of vertices, where the quality of an embedding is measured in terms of its load, expansion, dilation, and clocked congestion. Before we state our results, we introduce some basic definitions and notations.

For graph-theoretic terminology used but not explained here, we refer to [5]. Graphs, as well as digraphs, may have loops and multiple edges. For a graph (or

a digraph) $G$, $V(G)$ denotes its vertex set; $E(G)$ denotes the edge set of a graph $G$ (or the arc set of $G$ if $G$ is a digraph). For a graph, if there is just one edge joining the vertices $v$ and $w$, we denote this edge by $vw$. In a similar way, we use the symbols $(v, w)$ (or, alternatively, $v \longrightarrow w$) for digraphs. As usual, a *walk* in a graph $G$ is a sequence $(v_0, e_1, v_1, \ldots, e_k, v_k)$ whose terms are alternately vertices and edges of $G$, such that, for $1 \leq i \leq k$, the ends of $e_i$ are $v_{i-1}$ and $v_i$. A walk $(v_0, e_1, v_1, \ldots, e_k, v_k)$ is a *path* if $v_i \neq v_j$ for all $i \neq j$. Similarly, the notions of a *directed walk* and a *directed path* are defined. For a graph $G$, we denote by $G^*$ the digraph that results from $G$ in the following way. For each pair of distinct vertices $v, w$ of $G$ and each edge $e$ joining $v$ with $w$, choose a pair $\vec{e}_1, \vec{e}_2$ of arcs such that $v$ is the tail and $w$ is the head of $\vec{e}_1$ and, for the arc $\vec{e}_2$, $w$ is the tail and $v$ is the head. Then, assuming that all so chosen arcs are pairwise distinct, we define $G^*$ as the digraph that results from $G$ by replacing each edge $e$ joining distinct vertices by its corresponding arcs $\vec{e}_1, \vec{e}_2$. The construction of $G^*$ reflects the fact that the communication interfaces of most parallel computers of the MIMD (Multiple Instruction Multiple Data) type are bidirectional. The symbol $\mathbb{Z}_r$ denotes the integers modulo $r$ and operations of elements of $\mathbb{Z}_r$ are to be taken modulo $r$; we put $\mathbb{Z}_r^n := \{(x_{n-1}, \ldots, x_0) : x_i \in \mathbb{Z}_r (i = 0, \ldots, n-1)\}$.

For integers $d, D \geq 2$, the *directed de Bruijn graph* $\vec{B}(d, D)$ has as its vertex set the set $\mathbb{Z}_d^D$ and there exists an arc from $(a_{D-1}, \ldots, a_0)$ to $(b_{D-1}, \ldots, b_0)$ if and only if $b_i = a_{i-1} (i = 1, \ldots, D-1)$. The corresponding *undirected de Bruijn graph* $B(d, D)$ results from $\vec{B}(d, D)$ by ignoring the orientations of the arcs. We stress the fact that $B(d, D)$ contains loops and pairs of parallel edges. If $G = B(d, D)$, then $B^*(d, D)$ denotes the above defined digraph $G^*$. In the obvious way, we consider $\vec{B}(d, D)$ as a subgraph of $B^*(d, D)$. Let $\vec{e}$ be an arc of $B^*(d, D)$ joining distinct vertices. Then we say that $\vec{e}$ is in *shuffle direction* if $\vec{e} \in \vec{B}(d, D)$ and, otherwise, $\vec{e}$ is in *unshuffle direction*. Further, by definition, loops of $B^*(d, D)$ are both in shuffle and in unshuffle direction.

For $m \geq 3$, the *(2-dimensional quadratic) torus* $T(m)$ is the Cartesian product of two cycles of length $m$, i.e., $V(T(m)) = \mathbb{Z}_m \times \mathbb{Z}_m$ and two vertices $(a, b), (c, d)$ are joined by an edge iff either $a = c$ and $b, d$ differ by one or $b = d$ and $a, c$ differ by one (mod $m$).

Let $G, H$ be graphs and assume that $G$ is loopless. Let, further, $f : V(G) \longrightarrow V(H)$ be an injective mapping together with a mapping $P_f$ (called *routing*) which assigns to every arc $\vec{e} \in E(G^*)$ with tail $v$ and head $w$ a directed walk $P_f(\vec{e}) = (f(v) = v_0, \vec{e}_1, v_1, \ldots, \vec{e}_k, v_k = f(w))$ of $H^*$. Then the pair $(f, P_f)$ (and sometimes just the mapping $f$) is called an *embedding of $G$ into $H$*, and $\vec{e}_i$ is called the *$i$-th arc of $P_f(\vec{e})$* for $P_f(\vec{e}) = \big(f(v) = v_0, e_1, v_1, \ldots, e_k, v_k = f(w)\big)$. Further, $G$ is the *guest graph* and $H$ is the *host graph* of the embedding $(f, P_f)$. For a given embedding $(f, P_f)$ of $G$ into $H$ and for $\vec{h} \in E(H^*)$, let $\mu_i'(\vec{h})$ denote the number of arcs $\vec{e} \in E(G^*)$ for which $\vec{h}$ is the $i$-th arc of $P_f(\vec{e})$; put $\mu_i(\vec{h}) = \mu_i'(\vec{h})$ if $\vec{h}$ joins distinct vertices, and $\mu_i(\vec{h}) = \lceil \mu_i'(\vec{h})/2 \rceil$ otherwise. Then we define the *clocked congestion* of $(f, P_f)$ by

$$\text{clocked cong } (f, P_f) := \max\{\mu_i(\vec{h})\},$$

where the maximum is taken over all $\vec{h} \in E(H^*)$ and all $i \in \{1, \ldots, r\}$ with $r$ denoting the maximum length of the walks $P_f(\vec{e})$. A similar notion, called *dynamic congestion*, is considered in [2].

In the following, the walks $P_f(\vec{e})$ are called the *communication paths* of $(f, P_f)$. Informally speaking, the clocked congestion of an embedding is the maximum number of communication paths that, at the same time and in the same direction, pass through any edge $h$ of the host graph $H$, where we imagine that loops of $H$ can be traversed in two different directions. (In the above definition, the latter is reflected by putting $\mu_i(\vec{h}) = \lceil \mu_i'(\vec{h})/2 \rceil$ for loops $\vec{h}$.) In comparison with the commonly used measure of congestion, the definition of the clocked congestion provides a more realistic model of the communication capabilities of MIMD parallel computers since, in contrast to the usual notion of congestion, different communication paths can traverse the same arc at different points in time without conflict. This reflects the impact of message passing schemes.

The following definitions are standard. Let $(f, P_f)$ be an embedding of $G$ into $H$. Then the *dilation* of $(f, P_f)$ (which is denoted by $\mathrm{dil}(f, P_f)$) is the maximum length of the communication paths of $(f, P_f)$. The *load* of a mapping $f : V(G) \longrightarrow V(H)$ is the maximum number of vertices of $G$ which are mapped to the same vertex of $H$, and thus the fact that we consider only embeddings $(f, P_f)$ with injective $f$ can be expressed by saying that we restrict ourselves to embeddings with load 1 (in symbols, load $(f, P_f) = 1$). Finally, the *expansion* of $(f, P_f)$ is defined by putting expansion $(f, P_f) := \frac{|V(H)|}{|V(G)|}$.

For integers $m, d, D$ with $m \geq 3, d \geq 2$, and $D \geq 2$, let $T(m)$ and $B(d, D)$ be a torus and a de Bruijn graph, respectively, with the same number of vertices, i.e., $m^2 = d^D$. Assume that $m$ and $D$ are even and put $n = \frac{D}{2}$. Then $m = d^n$. Let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be a bijective mapping and let the mappings $\varphi_i : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d (i = 0, \ldots, n-1)$ be defined such that $\varphi(x) = (\varphi_{n-1}(x), \ldots, \varphi_0(x))$. We then obtain a bijective mapping $f_\varphi : V(T_m) \longrightarrow V(B(d, D))$ by putting (for $a, b \in \mathbb{Z}_m$)

$$f_\varphi\big((a, b)\big) = \begin{cases} (\varphi_{n-1}(a), \ldots, \varphi_0(a), \varphi_{n-1}(b), \ldots, \varphi_0(b)) \text{ if } a \equiv b \pmod 2 \\ (\varphi_{n-1}(b), \ldots, \varphi_0(b), \varphi_{n-1}(a), \ldots, \varphi_0(a)) \text{ otherwise .} \end{cases} \quad (1)$$

Then, based on the assumption that $m$ is even, one can define (in an obvious way) a corresponding routing $P_{f_\varphi}$ such that $\mathrm{dil}(f_\varphi, P_{f_\varphi}) = \frac{D}{2}$, load $(f_\varphi, P_{f_\varphi}) = 1$, and expansion $(f_\varphi, P_{f_\varphi}) = 1$. (For the details concerning $P_{f_\varphi}$, we refer to Section 2.)

For one particular choice of $\varphi$, namely, for the case in which $\varphi$ is the mapping $\kappa$ which corresponds to the base $d$ representation of the members of $\mathbb{Z}_{d^n}$ (see below), the mapping $f_\varphi$ was previously considered in [1, 6, 7] and has turned out to be one of the basic tools for constructing embeddings of other network topologies into de Bruijn graphs. One of our results states that clocked cong $(f_\varphi, P_{f_\varphi}) = 2$ for this particular choice of $\varphi$, and thus it is natural to ask whether or not there exist mappings $\varphi$ such that

$$\text{clocked cong } (f_\varphi, P_{f_\varphi}) = 1.$$

It is useful to introduce some notational conventions. For a mapping $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$, we throughout use the symbols $\varphi_i (i = 0, \ldots, n-1)$ as in the paragraph before (1). We write $\varphi = (\varphi_{n-1}, \ldots, \varphi_0)$. Let $\kappa = (\kappa_{n-1}, \ldots, \kappa_0)$ be the mapping $\kappa : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ which corresponds to the base $d$ representation of the members of $\mathbb{Z}_{d^n}$, i.e., for $a \in \mathbb{Z}_{d^n}$ the numbers $\kappa_i(a)$ are the uniquely determined elements of $\mathbb{Z}_d$ for which $a = \sum_{i=0}^{n-1} \kappa_i(a) d^i$. We use the following convention: whenever one of the symbols $a, b, a', b', x, y$ denotes a member of $\mathbb{Z}_{d^n}$, then we write $\alpha_i, \beta_i, \alpha_i', \beta_i', \xi_i$, and $\eta_i$ for $\kappa_i(a), \kappa_i(b), \kappa_i(a'), \kappa_i(b')$, $\kappa_i(x)$, and $\kappa_i(y)$, respectively.

We now describe our main results. Among the mappings $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$, we consider mappings of a certain type which are defined as follows. For all integers $i, j$ with $0 \leq j \leq i \leq n-1$, let $\lambda_{i,j} \in \mathbb{Z}_d$ and let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$, $\varphi = (\varphi_{n-1}, \ldots, \varphi_0)$, be defined by

$$\varphi_i(a) = \sum_{j=0}^{i} \lambda_{i,j} \alpha_j \quad (a \in \mathbb{Z}_{d^n},\ i = 0, \ldots, n-1). \tag{2}$$

Then $\varphi$ is called a *mapping of triangular type* and the $\lambda_{i,j} (0 \leq j \leq i \leq n-1)$ are its *coefficients*. We note that the operations in (2) are to be taken modulo $d$ (because the operands are in $\mathbb{Z}_d$ and because we want the $\varphi_i(a)$ to be in $\mathbb{Z}_d$). In a similar manner, whenever the meaning is clear from the context, several other of the identities and operations in this paper are understood to be mod $d$ or mod $d^n$ without explicit mention.

For $d > 2$, our main result characterizes, in terms of the coefficients $\lambda_{i,j}$, the bijective mappings $\varphi$ of triangular type for which clocked cong $(f_\varphi, P_{f_\varphi})$ is equal to 1. As a consequence, for $d > 2$, one obtains that there exist bijective mappings $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ of triangular type such that clocked cong $(f_\varphi, P_{f_\varphi}) = 1$.

This solves the above mentioned existence problem for the case in which $d$ is greater than two. For $d = 2$, we do not have a complete solution of this problem, but have obtained the following partial results:

(i) there exist no bijective mappings $\varphi : \mathbb{Z}_{2^n} \longrightarrow \mathbb{Z}_2^n$ of triangular type for which clocked cong $(f_\varphi, P_{f_\varphi})$ is equal to 1,
(ii) for $n = 2$, bijective mappings $\varphi : \mathbb{Z}_{2^n} \longrightarrow \mathbb{Z}_2^n$ with clocked cong $(f_\varphi, P_{f_\varphi}) = 1$ exist, and
(iii) no such mapping exists for $n = 3$.

## 2 The clocked congestion one condition

Let $m, d, D, n$, and $\varphi = (\varphi_{n-1}, \ldots, \varphi_0)$ be as in the paragraph preceding (1) and let $f_\varphi : V(T(m)) \longrightarrow V\left(B(d, D)\right)$ be the bijective mapping defined by (1). We want to define a corresponding routing $P_{f_\varphi}$. For this purpose, let $e \in E\left(T(m)\right)$. Then there are $a, a', x \in \mathbb{Z}_{d^n}$ such that $a$ and $a'$ differ by one (mod $d^n$) and such that either $e = (a, x)(a', x)$ or $e = (x, a)(x, a')$. Note that, since $m$ is even, we have $a \not\equiv a' \pmod 2$. If $e = (a, x)(a', x)$, then (by symmetry) we may assume

$a \equiv x \pmod 2$, and we put $v = (a, x), w = (a', x)$; if $e = (x, a)(x, a')$, then we assume $a' \equiv x \pmod 2$ and put $v = (x, a), w = (x, a')$. In either case, one concludes from $a \not\equiv a' \pmod 2$ that

$$f_\varphi(v) = \big(\varphi_{n-1}(a), \ldots, \varphi_0(a), \varphi_{n-1}(x), \ldots, \varphi_0(x)\big),$$
$$f_\varphi(w) = \big(\varphi_{n-1}(x), \ldots, \varphi_0(x), \varphi_{n-1}(a'), \ldots, \varphi_0(a')\big).$$

We now define $P_{f_\varphi}$. (In the following, for notational simplicity, we write $P$ instead of $P_{f_\varphi}$.) We specify directed walks ("communication paths") $P(v \to w)$ and $P(w \to v)$ in $B^*(d, D)$ as follows. Let $P(v \to w) = (v_0, \vec{e}_1, v_1, \ldots, \vec{e}_n, v_n)$ with

$$v_i = \begin{cases} f_\varphi(v) \text{ for } i = 0 \\ f_\varphi(w) \text{ for } i = n \\ \big(\varphi_{n-1-i}(a), \ldots, \varphi_0(a), \varphi_{n-1}(x), \ldots, \varphi_0(x), \varphi_{n-1}(a'), \ldots, \varphi_{n-i}(a')\big) \\ \text{for } 1 \le i \le n-1 \end{cases}$$

and such that all arcs $\vec{e}_i$ are in shuffle direction. (We express this by saying that $P(v \to w)$ *is in shuffle direction.*) Let further $P(w \to v) = (v_n, \vec{f}_n, v_{n-1}, \ldots, \vec{f}_1, v_0)$, where $P(w \to v)$ traverses the same vertices $v_i$ as $P(v \to w)$, but in opposite order, and where $\vec{f}_i$ is the arc of $B^*(d, D)$ in unshuffle direction corresponding to $\vec{e}_i$ $(i = 1, \ldots, n)$. This defines $P = P_{f_\varphi}$. Finally, let $\mathcal{P}$ denote the set of all so defined communication paths $P(v \to w)$ which are in shuffle direction. We claim that the statement

$$\text{clocked cong}(f_\varphi, P) = 1 \tag{3}$$

is equivalent to the following condition (4), which we call the *clocked congestion one condition* (*cc1-condition* for short).

For all $a, a', b, b' \in \mathbb{Z}_{d^n}$ such that $(a, a') \neq (b, b')$, $a' = a - 1$ or $a + 1$, $b' = b - 1$ or $b + 1$ and for all $i \in \{0, \ldots, n-1\}$, the $(n+1)$-tuples $(\varphi_i(a), \ldots, \varphi_0(a), \varphi_{n-1}(a'), \ldots, \varphi_i(a'))$ and $\big(\varphi_i(b), \ldots, \varphi_0(b), \varphi_{n-1}(b'), \ldots, \varphi_i(b')\big)$ are distinct. (4)

We first show that (4) implies (3). Hence, assuming that (4) holds, let $e = vw, \tilde{e} = \tilde{v}\tilde{w}$ be distinct edges of $T(m)$, where the notations are chosen such that $P(v \to w)$ and $P(\tilde{v} \to \tilde{w})$ are in $\mathcal{P}$. For the edge $e$, we use the symbols $a, a', x$ as in the first paragraph of this section. Analogously, for the edge $\tilde{e}$, the symbols $b, b', y$ are used.

Then $a \equiv x \pmod 2$ if $e = (a, x)(a', x)$, and $a' \equiv x \pmod 2$ if $e = (x, a)(x, a')$. Similarly, $b \equiv y \pmod 2$ if $\tilde{e} = (b, y)(b', y)$, and $b' \equiv y \pmod 2$ if $\tilde{e} = (y, b)(y, b')$. From this, together with $e \neq \tilde{e}$ and $a \not\equiv a' \pmod 2$, one easily finds that $x = y$ implies $(a, a') \neq (b, b')$. Further, one easily obtains that, in order to prove that (3) holds, it is sufficient to show that

$$\big(\varphi_i(a), \ldots, \varphi_0(a), \varphi_{n-1}(x), \ldots, \varphi_0(x), \varphi_{n-1}(a'), \ldots, \varphi_i(a')\big) \neq \big(\varphi_i(b), \ldots, \varphi_0(b), \varphi_{n-1}(y), \ldots, \varphi_0(y), \varphi_{n-1}(b'), \ldots, \varphi_i(b')\big) \tag{5}$$

for all $i \in \{0, \ldots, n-1\}$.

For $x \neq y$, this clearly holds, and thus we can assume $x = y$. Hence $(a, a') \neq (b, b')$, and we can use (4) to obtain (5).

In order to show that (4) follows from (3), suppose that (3) holds, but (4) does not hold. Then there are $a, a', b, b' \in \mathbb{Z}_{d^n}, i \in \{0, 1, \ldots, n-1\}$ with $(a, a') \neq (b, b')$, $a' = a + 1$ or $a - 1$, $b' = b + 1$ or $b - 1$ such that

$$\begin{aligned}&\big(\varphi_i(a), \ldots, \varphi_0(a), \varphi_{n-1}(a'), \ldots, \varphi_i(a')\big) = \\ &\big(\varphi_i(b), \ldots, \varphi_0(b), \varphi_{n-1}(b'), \ldots, \varphi_i(b')\big).\end{aligned} \tag{6}$$

Pick $z \in \mathbb{Z}_{d^n}$ with $z \equiv a \pmod 2$ and let $P_1 := P\big((a, z) \to (a', z)\big)$. Let $P_2 := P\big((b, z) \to (b', z)\big)$ if $b \equiv z \pmod 2$ and $P_2 := P\big((z, b) \to (z, b')\big)$ otherwise. Then $P_1, P_2 \in \mathcal{P}$. Let further

$$\begin{aligned}w_1 &:= \big(\varphi_i(a), \ldots, \varphi_0(a), \varphi_{n-1}(z), \ldots, \varphi_0(z), \varphi_{n-1}(a'), \ldots, \varphi_{i+1}(a')\big) \quad \text{and} \\ w_2 &:= \big(\varphi_{i-1}(a), \ldots, \varphi_0(a), \varphi_{n-1}(z), \ldots, \varphi_0(z), \varphi_{n-1}(a'), \ldots, \varphi_i(a')\big).\end{aligned}$$

Then, by definition of $P_1, P_2$ and because of (6), the $(n - i)$-th arc of both $P_1$ and $P_2$ is the uniquely determined arc of $B^*(d, D)$ which has $w_1$ as its tail and $w_2$ as its head and which is in shuffle direction. Hence, because of (3), the arc $(a, z) \to (a', z)$ must be identical to one of the arcs $(b, z) \to (b', z)$, or $(z, b) \to (z, b')$ which in either case contradicts the fact that $(a, a') \neq (b, b')$. This shows that (4) follows from (3).

## 3 Main results

Throughout this section we assume that $d, n$ are integers with $d \geq 2$ and $n \geq 1$. We start with a simple lemma.

**Lemma.** *Let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be of triangular type with coefficients $\lambda_{i,j} (0 \leq j \leq i \leq n-1)$. For some $i \in \{0, \ldots, n-1\}$ and some $a, b \in \mathbb{Z}_{d^n}$, assume that $gcd(\lambda_{i,i}, d) = 1$, $\varphi_i(a) = \varphi_i(b)$, and $\alpha_j = \beta_j$ for all $j$ with $0 \leq j < i$. Then $\alpha_i = \beta_i$.*

*Proof.* From $\varphi_i(a) = \varphi_i(b)$ and $\alpha_j = \beta_j$ $(0 \leq j < i)$, together with (2), one obtains $\lambda_{i,i}\alpha_i = \lambda_{i,i}\beta_i$, and thus $\alpha_i = \beta_i$, because the computations are mod $d$ and we are assuming $gcd(\lambda_{i,i}, d) = 1$. □

Based on this lemma, we show the following.

**Proposition 1.** *Let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be of triangular type with coefficients $\lambda_{i,j} (0 \leq j \leq i \leq n-1)$. Then $\varphi$ is bijective if and only if $gcd(\lambda_{i,i}, d) = 1$ for all $i \in \{0, \ldots, n-1\}$.*

*Proof.* Assume first that $gcd(\lambda_{i,i}, d) = 1 (i = 0, \ldots, n-1)$ and let $a, b \in \mathbb{Z}_{d^n}$ with $\varphi(a) = \varphi(b)$. Then, by recursive application of the lemma, $\alpha_i = \beta_i \, (i = 0, \ldots, n-1)$. Hence $a = b$, and thus the "if direction" is settled.

For the proof of the converse, assume $gcd(\lambda_{i,i}, d) = t > 1$ for some $i$. Let $A = \{a \in \mathbb{Z}_{d^n} : \alpha_j = 0 \text{ for } j = 0, \ldots, i-1, \alpha_i = 0 \text{ or } \frac{d}{t}\}$. Then it follows from (2), together with $\lambda_{i,i}\frac{d}{t} \equiv 0 \pmod{d}$, that $\varphi_j(a) = 0$ for all $a \in A$ and all $j \in \{0, \ldots, i\}$. Hence $|\varphi(A)| \leq d^{n-i-1}$. On the other hand, we have $|A| = 2d^{n-i-1}$, and thus $\varphi$ is not bijective. □

**Proposition 2.** *Let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be of triangular type with coefficients $\lambda_{i,j}$ $(0 \leq j \leq i \leq n-1)$ and assume that $\varphi$ satisfies the cc1-condition. Let further $r, s$ be integers with $0 \leq s \leq r \leq n-1$ and assume that $d > 2$ if $s = 0$. Then*

$$\lambda_{r,0} + \ldots + \lambda_{r,s} \neq -\lambda_{r,0}. \tag{7}$$

*Proof.* Suppose that (7) does not hold. Then

$$\lambda_{r,s} = \lambda_{r,s-1}(d-1) + \ldots + \lambda_{r,1}(d-1) + \lambda_{r,0}(d-2) \text{ if } s > 0 \tag{8}$$

and

$$2\lambda_{r,0} = 0 \text{ if } s = 0. \tag{9}$$

Let $A = \{a \in \mathbb{Z}_{d^n} : \alpha_j = d-1 \, (0 \leq j \leq s-1), \alpha_j = 0 \, (s \leq j \leq r)\}$ if $s > 0$ and $A = \{a \in \mathbb{Z}_{d^n} : \alpha_0 = 1, \alpha_j = 0 \, (1 \leq j \leq r)\}$, otherwise. Put $A' = \{a' \in \mathbb{Z}_{d^n} : a' = a-1 \text{ or } a+1 \text{ for some } a \in A\}$ and let $\mu = \lambda_{r,s}$ if $s > 0$ and $\mu = 0$, otherwise. Then it follows from (2), (8), and (9), together with the assumption that $d > 2$ if $s = 0$, that $\varphi_r(a') = \mu$ for all $a' \in A'$. Moreover, it follows from the definition of $A$, together with (2), that $\varphi_j(a) = \varphi_j(b)$ for all $a, b \in A$ and all $j$ with $0 \leq j \leq r$. Hence

$$|\{(\varphi_r(a), \ldots, \varphi_0(a), \varphi_{n-1}(a'), \ldots, \varphi_r(a')) : a \in A, a' = a-1 \text{ or } a+1\}| \leq d^{n-1-r}.$$

On the other hand, there are $2d^{n-1-r}$ distinct pairs $(a, a')$ with $a \in A$, $a' = a-1$ or $a+1$, and thus we have obtained a contradiction to the assumption that $\varphi$ satisfies the cc1-condition. □

The next theorem, its corollary and the subsequent Theorem 6 are the main results of this paper.

**Theorem 3.** *For $d > 2$, let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be of triangular type with coefficients $\lambda_{i,j}$ $(0 \leq j \leq i \leq n-1)$. Then $\varphi$ is a bijective mapping satisfying the cc1-condition if and only if the following statements (i) and (ii) hold:*

*(i)* $gcd(\lambda_{i,i}, d) = 1$ *for all* $i \in \{0, \ldots, n-1\}$
*(ii)* $\lambda_{i,0} + \ldots + \lambda_{i,j} \neq -\lambda_{i,0}$ *for all* $i, j$ *with* $0 \leq j \leq i \leq n-1$.

*Proof.* The propositions 1, 2 already cover the "only if" direction. For the converse direction, assume that (i) and (ii) hold. Then, by Proposition 1, $\varphi$ is bijective, and thus it remains to show that $\varphi$ satisfies the *cc*1-condition. To this end, suppose that there exist $a, a', b, b' \in \mathbb{Z}_{d^n}$ with $(a, a') \neq (b, b'), a' = a - 1$ or $a + 1$, $b' = b - 1$ or $b + 1$ such that, for some $i \in \{0, \ldots, n-1\}$,

$$\varphi_j(a) = \varphi_j(b) \ (0 \leq j \leq i) \text{ and } \varphi_j(a') = \varphi_j(b') \ (i \leq j \leq n-1). \tag{10}$$

By recursive application of the lemma, it follows from (10) that

$$\alpha_j = \beta_j \ (0 \leq j \leq i). \tag{11}$$

We separately discuss two cases.
*Case 1.* $a' = a - 1, b' = b - 1$ or $a' = a + 1, b' = b + 1$.
Then it follows from (11), together with the hypothesis of case 1, that $\alpha'_j = \beta'_j \ (j = 0, \ldots, i)$. Consequently, by (10) and recursive application of the lemma, we have $\alpha'_j = \beta'_j \ (j = i+1, \ldots, n-1)$. Hence $a' = b'$, and thus $a = b$, in contradiction to the assumption $(a, a') \neq (b, b')$. This settles case 1 and thus, by symmetry, it remains to consider the following case.
*Case 2.* $a' = a + 1, b' = b - 1$.
Then, since $\alpha_0 = \beta_0$ by (11), there exists some $k \in \{0, \ldots, i\}$ such that $\alpha'_j = \alpha_j + 1 \ (0 \leq j \leq k), \alpha'_j = \alpha_j \ (k < j \leq i), \beta'_0 = \beta_0 - 1, \beta'_j = \beta_j \ (1 \leq j \leq i)$, or such that $\beta'_j = \beta_j - 1 \ (0 \leq j \leq k), \beta'_j = \beta_j \ (k < j \leq i), \alpha'_0 = \alpha_0 + 1, \alpha'_j = \alpha_j \ (1 \leq j \leq i)$. In either case, (11) implies $\alpha'_0 - \beta'_0 = 2, \alpha'_j - \beta'_j = 1 \ (1 \leq j \leq k), \alpha'_j - \beta'_j = 0 (k < j \leq i)$. Further, by (10),

$$\lambda_{i,0}\alpha'_0 + \ldots + \lambda_{i,i}\alpha'_i = \varphi_i(a') = \varphi_i(b') = \lambda_{i,0}\beta'_0 + \ldots + \lambda_{i,i}\beta'_i$$

and thus

$$\lambda_{i,1}(\alpha'_1 - \beta'_1) + \ldots + \lambda_{i,i}(\alpha'_i - \beta'_i) = \lambda_{i,0}(\beta'_0 - \alpha'_0)$$

where the left hand side is understood to be zero if $i = 0$. It follows that $\lambda_{i,1} + \ldots + \lambda_{i,k} = -2\lambda_{i,0}$ if $k \geq 1$ and $0 = -2\lambda_{i,0}$, otherwise, which contradicts the assumption that (ii) holds. □

As a consequence of Theorem 3, one obtains the following corollary.

**Corollary 4.** *For all $d, n$ with $d > 2$ and $n \geq 1$, there exist bijective mappings $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ of triangular type satisfying the cc1-condition.*

*Proof.* Let $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ be of triangular type with coefficients $\lambda_{i,0} = 1 (i = 0, \ldots, n-1), \lambda_{i,i} = -1 \ (i = 1, \ldots, n-1)$, and $\lambda_{i,j} = 0$, otherwise. Then $\varphi$ satisfies the conditions (i) and (ii) of Theorem 3. (For other examples, see below.) Hence the assertion follows from Theorem 3. □

Given a bijective mapping $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n$ satisfying the *cc*1-condition, the next proposition shows how to obtain from $\varphi$ other mappings $\varphi'$ having the same nice properties.

**Proposition 5.** *If $\varphi : \mathbb{Z}_{d^n} \longrightarrow \mathbb{Z}_d^n, \varphi = (\varphi_{n-1}, \ldots, \varphi_0)$, is bijective and satisfies the cc1-condition, then the same holds for $\varphi' = (\varphi'_{n-1}, \ldots, \varphi'_0)$ with $\varphi'_i = \sigma_i \circ \varphi_i$ where the $\sigma_i$ are permutations of $\mathbb{Z}_d$ $(i = 0, 1, \ldots, n-1)$.*

*Proof.* Clearly $\varphi'$ is bijective. Assume that $\varphi'$ does not satisfy the cc1-condition. Then there exists an $i, 0 \leq i < n$, together with $a, b \in \mathbb{Z}_{d^n}$ such that $(a, a') \neq (b, b')$, $a' = a + 1$ or $a - 1, b' = b + 1$ or $b - 1$ and such that

$$(\varphi'_i(a), \ldots, \varphi'_0(a), \varphi'_{n-1}(a'), \ldots, \varphi'_i(a')) = (\varphi'_i(b), \ldots, \varphi'_0(b), \varphi_{n-1}(b'), \ldots, \varphi_i(b')).$$

Hence

$$\sigma_j \circ \varphi_j(a) = \sigma_j \circ \varphi_j(b) \quad \text{for all } j = 0, 1, \ldots, i \quad \text{and}$$
$$\sigma_j \circ \varphi_j(a') = \sigma_j \circ \varphi_j(b') \quad \text{for all } j = i, \ldots, n-1.$$

Multiplying these equations with $\sigma_j^{-1}$ from the left side yields $\varphi_j(a) = \varphi_j(b)$ for all $j = 0, 1, \ldots, i$ and $\varphi_j(a') = \varphi_j(b')$ for all $j = i, \ldots, n-1$. This means that $\varphi$ violates the cc1-condition, in contradiction to the assumptions on $\varphi$. □

By making use of Theorem 3, it is easy to construct triangular type mappings $\varphi$ which are bijective and satisfy the cc1-condition (for given values of $d$ and $n$ such that $d \geq 3$ and $n \geq 1$). Having already fixed the values of $\lambda_{0,0}, \lambda_{1,0}, \lambda_{1,1}, \lambda_{2,0}, \ldots, \lambda_{k,0}, \ldots, \lambda_{k,l}$, the choice of the next candidate $\lambda_{k',l'}$ with $(k', l') = (k, l+1)$ if $l < k$, and $(k', l') = (k+1, 0)$ otherwise, is (according to Theorem 3) only limited by

(i) $gcd(\lambda_{k',l'}, d) = 1$ if $k' = l'$

and (ii) $\lambda_{k',0} + \ldots + \lambda_{k',l'} \neq -\lambda_{k',0}$.

For example the following list of coefficients for $n = 3$, $d = 3$ satisfies this restriction (at each step the forbidden values are given in parentheses):

$$\lambda_{0,0} = 2(0),\ \lambda_{1,0} = 1(0),\ \lambda_{1,1} = 2(0,1),\ \lambda_{2,0} = 2(0),\ \lambda_{2,1} = 0(2),\ \lambda_{2,2} = 1(0,2).$$

**Theorem 6.** *For $n \geq 2$, there exist no bijective mappings $\varphi : \mathbb{Z}_{2^n} \longrightarrow \mathbb{Z}_2^n$ of triangular type satisfying the cc1-condition.*

*Proof.* Let $n \geq 2$ and suppose that $\varphi : \mathbb{Z}_{2^n} \longrightarrow \mathbb{Z}_2^n$ is a bijective mapping of triangular type such that $\varphi$ satisfies the cc1-condition. Let $\lambda_{i,j} (0 \leq j \leq i \leq n-1)$ denote the coefficients of $\varphi$. We claim that

$$\lambda_{i,j} = 0 \text{ for all } i, j \text{ with } 1 \leq j < i \leq n-1. \tag{12}$$

For the proof of (12), suppose that $\lambda_{i,j} = 1$ for some indices $i, j$ with $1 \leq j < i \leq n-1$ and assume that $j$ is chosen such that $\lambda_{i,k} = 0$ for all $k$ with $1 \leq k < j$. Let $h$ be the least index greater than $j$ such that $\lambda_{i,h} = 1$. (An index $h$ with these properties exists because $\lambda_{i,i} = 1$ by Proposition 1.) But then

$$\lambda_{i,0} + \ldots + \lambda_{i,h} = \lambda_{i,0} = -\lambda_{i,0},$$

which (because $h > 0$) contradicts Proposition 2. This proves (12).

As a consequence of (2), (12), and the fact that $\lambda_{i,i} = 1$ by Proposition 1, one obtains that

$$\varphi_0(x) = \xi_0,\ \varphi_i(x) = \lambda_{i,0}\xi_0 + \xi_i\ (1 \leq i \leq n-1) \tag{13}$$

for all $x \in \mathbb{Z}_{2^n}$. Now, by means of (13), it is easy to find $a, a', b, b' \in \mathbb{Z}_{2^n}$ with $(a, a') \neq (b, b'), a' = a - 1$ or $a + 1, b' = b - 1$ or $b + 1$ such that, for some $i \in \{0, \ldots, n-1\}$, the *cc*1-condition is violated. We just specify the choices of $a, a', b, b'$ and the value of $i$, leaving all further details to the reader:

$$\begin{array}{l} a = 1,\ b = 1,\ a' = 0,\ b' = 2,\ i = 2 \text{ if } n \geq 3 \\ a = 1,\ b = 3,\ a' = 0,\ b' = 0,\ i = 0 \text{ if } n = 2,\ \lambda_{1,0} = 0 \\ a = 0,\ b = 2,\ a' = 1,\ b' = 1,\ i = 0 \text{ otherwise .} \end{array}$$

□

Theorem 6 settles the case of mappings of triangular type. The general problem whether or not there exist bijective mappings $\varphi : \mathbb{Z}_{2^n} \longrightarrow \mathbb{Z}_2^n$ satisfying the *cc*1-condition remains open. For $n = 1$ and $n = 2$ mappings $\varphi$ with these properties exist: for $n = 1$, we just choose the identity mapping, while for $n = 2$, the mapping $\varphi$ defined by $\varphi(a) = (\alpha_1, \alpha_1 + \alpha_0)$ has the desired properties, as can easily be checked. For $n = 3$, however, one can show that no such mappings $\varphi$ exist: the proof can be carried out by exhaustive case distinctions, where by symmetry it suffices to consider a few cases. We leave the details to the reader.

## 4 An embedding with clocked congestion two

Among the mappings $f_\varphi : V(T(m)) \to V(B(d, D))$ defined as in (1), probably the first example which comes to mind is the mapping which corresponds to the base $d$ representation of the members of $\mathbb{Z}_{d^n}$, i.e. the mapping $f_\kappa$ where $\kappa : \mathbb{Z}_{d^n} \to \mathbb{Z}_d^n$ is defined as in the introduction. We have already mentioned that the mapping $f_\kappa$ was previously considered in [1, 6, 7]. Observe that Theorem 3 immediately implies clocked cong $(f_\kappa, P_{f_\kappa}) \geq 2$. In this section, we prove the following result.

**Theorem 7.** *Let $\kappa : \mathbb{Z}_{d^n} \to \mathbb{Z}_d^n$ be defined as in the introduction. Then clocked cong $(f_\kappa, P_{f_\kappa}) = 2$.*

We first prove the subsequent Proposition 8 and then obtain Theorem 7 as a consequence of this proposition.

**Proposition 8.** *Let $a, a', b, b' \in \mathbb{Z}_{d^n}$ be such that $(a, a') \neq (b, b'), a' = a - 1$ or $a + 1, b' = b - 1$ or $b + 1$ and assume that, for some $i \in \{0, \ldots, n-1\}$, $(\alpha_i, \ldots, \alpha_0, \alpha'_{n-1}, \ldots, \alpha'_i) = (\beta_i, \ldots, \beta_0, \beta'_{n-1}, \ldots, \beta'_i)$. Then either $a = b$ or $a' = b'$, where the latter can occur only if $d = 2$.*

*Proof.* By assumption, we have

$$\alpha_j = \beta_j \ (j = 0, \ldots, i), \tag{14}$$
$$\alpha'_j = \beta'_j \ (j = i, \ldots, n-1). \tag{15}$$

We show that $a \neq b$ implies $a' = b'$ and $d = 2$. Hence let $a \neq b$. Then, because of (14), there exists an index $k, i < k \leq n-1$, such that $\alpha_k \neq \beta_k$. By (15), we have $\alpha'_k = \beta'_k$, and thus $\alpha_k \neq \alpha'_k$ or $\beta_k \neq \beta'_k$. Clearly, it suffices to settle the case in which the former holds, which we assume now. From this, together with $a' = a - 1$ or $a + 1$, we conclude $\alpha_j \neq \alpha'_j (j = 0, \ldots, k)$. Hence $\alpha_i \neq \alpha'_i$, which together with (14) and (15) implies $\beta_i \neq \beta'_i$. Hence, because $b' = b - 1$ or $b + 1$, we have $\beta_j \neq \beta'_j (j = 0, \ldots, i)$. Thus we have

$$\alpha_j \neq \alpha'_j, \ \beta_j \neq \beta'_j \ (j = 0, \ldots, i). \tag{16}$$

If $a' = a - 1, b' = b - 1$ or $a' = a + 1, b' = b + 1$, then (14) implies $\alpha'_j = \beta'_j \ (j = 0, \ldots, i)$, from which, together with (15), one concludes $a' = b'$. Consequently, we have $a = b$, in contradiction to the hypothesis $a \neq b$. Hence

$$a' = a - 1, \ b' = b + 1 \text{ or } a' = a + 1, \ b' = b - 1. \tag{17}$$

From (14), (16), and (17) one concludes $i = 0$. Hence $a' = b'$ by (15). Moreover, one concludes from $i = 0$, together with (14) and (15), that $\alpha_0 = \beta_0, \alpha'_0 = \beta'_0$. But this is compatible with (17) only if $d = 2$, and thus we have finished the proof of Proposition 8. □

*Proof of Theorem 7.* It remains to show clocked cong $(f_\kappa, P_{f_\kappa}) \leq 2$. For this purpose, let $e = vw, \tilde{e} = \tilde{v}\tilde{w}$ be edges of $T(m)$, where the notations are chosen such that $P(v \to w), P(\tilde{v} \to \tilde{w}) \in \mathcal{P}$. (For simplicity, we write $P$ instead of $P_{f_\kappa}$; for the definition of $\mathcal{P}$, see Section 2.) We use the symbols $a, a', x$ as in the first paragraph of Section 2. Analogously, for the edge $\tilde{e}$, the symbols $b, b', y$ are used. Assume that the communication paths $P(v \to w)$, $P(\tilde{v} \to \tilde{w})$ have their $(n-i)$-th arc in common. Then

$$(\alpha_i, \ldots, \alpha_0, \xi_{n-1}, \ldots, \xi_0, \alpha'_{n-1}, \ldots, \alpha'_i) = (\beta_i, \ldots, \beta_0, \eta_{n-1}, \ldots, \eta_0, \beta'_{n-1}, \ldots, \beta'_i).$$

Hence $x = y$. Further, Proposition 8 implies that $a = b$ or $a' = b'$. In each case, we have $a \equiv b \pmod 2$ and thus $a \equiv x \pmod 2$ if and only if $b \equiv y \pmod 2$. Hence we either have $e = (a, x)(a', x)$, $\tilde{e} = (b, y)(b', y)$ or $e = (x, a)(x, a')$, $\tilde{e} = (y, b)(y, b')$.

Now suppose that $\tilde{\tilde{e}}$ is an edge of $T(m)$ such that the corresponding communication path $P(\tilde{\tilde{v}} \to \tilde{\tilde{w}}) \in \mathcal{P}$ has the $(n-i)$-th arc in common with both $P(\tilde{v} \to \tilde{w})$ and $P(v \to w)$. For $\tilde{\tilde{e}}$, we use the symbols $c, c'$ analogously to the use of $a, a'$ and $b, b'$ for $e$ and $\tilde{e}$, respectively. We may assume that $e = (a, x)(a', x)$ because the case $e = (x, a)(x, a')$ can be treated similarly. Then $\tilde{e} = (b, x)(b', x)$, $\tilde{\tilde{e}} = (c, x)(c', x)$ with

$$a = b \text{ or } a' = b',$$
$$b = c \text{ or } b' = c',$$
$$a = c \text{ or } a' = c'.$$

Hence $a = b = c$ or $a' = b' = c'$, which implies (because $a' = a - 1$ or $a + 1$, $b' = b - 1$ or $b + 1$, $c' = c - 1$ or $c + 1$) that two of the edges $e, \widetilde{e}, \widetilde{\widetilde{e}}$ must be identical. This shows that clocked $\text{cong}(f_\kappa, P_{f_\kappa}) \leq 2$. □

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